A Three-Person Attendance Game with Unknown Infection Status

Definition 1 (Players, actions, and information)**.** *Three people all know one another. Each simultaneously chooses to attend (A) or stay home (H). Nobody observes anyone's infection status before choosing, including their own. Each person's initial infection status is independently drawn: they are infected with probability π and susceptible with probability $1-\pi$.*

Definition 2 (Transmission and costs)**.** *An infected attendee transmits infection to each susceptible attendee with probability β, independently across contacts. Staying home costs $c > 0$. Being infected, either initially or through a contact during this day, costs $L > 0$. This is a one-day game, so recovery is not included.*

Definition 3 (Expected payoff)**.** *Let $k \in \{0, 1, 2\}$ be the number of other players who attend. Write $U(A \mid k)$ and $U(H \mid k)$ for a player's expected payoffs from attending and staying home, respectively. Then*

$$U(A \mid k) = -L\left[\pi + (1-\pi)\left(1-(1-\pi\beta)^k\right)\right], \tag{1}$$

$$U(H \mid k) = -L\pi - c. \tag{2}$$

Derivation of the expected payoffs. The player is initially infected with probability π, which contributes an expected infection cost of $L\pi$ under either action.

Suppose the player is initially susceptible. Each of the k attending neighbors is infected with probability π. Conditional on being infected, that neighbor transmits with probability β. Hence one attending neighbor does not infect the player with probability $1-\pi\beta$. The probability of avoiding infection from all k neighbors is $(1-\pi\beta)^k$. The probability of a new infection is therefore $1-(1-\pi\beta)^k$. Multiplying by the probability $1-\pi$ that the player was initially susceptible gives the additional infection probability in (1).

A player who stays home has no workplace contacts, but pays c. This gives (2). □

Proposition 1 (Best responses and pure Nash equilibria)**.** *For $L = 10$, $\beta = 0.04$, $\pi = 0.1$, and $c = 0.05$, attending is optimal when zero or one other player attends. Staying home is optimal when both other players attend. The pure Nash equilibria are exactly*

$$(A, A, H), \qquad (A, H, A), \qquad (H, A, A).$$

Proof. Subtracting (2) from (1), the gain from attending is

$$\Delta(k) = U(A \mid k) - U(H \mid k) = c - L(1-\pi)\left[1-(1-\pi\beta)^k\right]. \tag{3}$$

Here $\pi\beta = 0.004$ and $L(1-\pi) = 9$. Thus

$$\Delta(0) = 0.05 - 9(1-0.996^0) = 0.05 > 0,$$
$$\Delta(1) = 0.05 - 9(1-0.996^1) = 0.014 > 0,$$
$$\Delta(2) = 0.05 - 9(1-0.996^2) = -0.021856 < 0.$$

At any outcome with exactly two attendees, each attendee faces one other attendee and prefers A. The player at home faces two attendees and prefers H. These three outcomes are therefore Nash equilibria. At an outcome with fewer than two attendees, a player at home prefers to switch to A. At (A, A, A), each player prefers to switch to H. There are no other pure Nash equilibria. □

Proposition 2 (A fair recommendation)**.** *A device that chooses each of the three pure Nash equilibria with probability $1/3$ and privately recommends each player's assigned action is a correlated equilibrium. Each player is recommended H with probability $1/3$ and A with probability $2/3$.*

Proof. Conditional on an A recommendation, exactly one other player is assigned A, so obeying yields gain $\Delta(1) = 0.014 > 0$ relative to switching to H. Conditional on an H recommendation, both other players are assigned A, so switching to A would yield $\Delta(2) = -0.021856 < 0$. Therefore no player benefits from deviating from either recommendation. □